



Data-Based Critical Thinking

STA1DCT

LMS Subject link

<https://lms.latrobe.edu.au/course/view.php?id=131132>

SUBJECT LEARNING GUIDE

Semester 1, 2023

Subject Coordinator: Dr David Farchione

SUBJECT DETAILS

Subject Code:	STA1DCT
Subject Title:	Data-Based Critical Thinking
Teaching Period:	Semester 1
Location(s):	Bundoora, Bendigo
Credit Points:	15
Delivery Mode:	Day
AQF Level:	Undergraduate

ENROLMENT REQUIREMENTS

Prerequisites:	None
Co-requisites:	None
Assumed Skills & Knowledge:	None
Special Study Requirements:	None

Additional subject requirements: STA1DCT is incompatible with STA2DCT (last offered in 2014)

STAFF CONTACTS

Subject Coordinator:	Dr David Farchione
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ABOUT THIS SUBJECT

This subject helps the student evaluate data-based evidence encountered in everyday life. It provides the fundamental numeracy skills required by business people, scientists, lawyers, nurses, journalists, social scientists, teachers and other professionals who need to evaluate data-based arguments, whether found in newspapers, television or on-line websites. This is achieved by a combination of studying newsworthy topics introduced in lectures, laboratory classes which encourage engagement with others and on-line quizzes that assess numeracy skills. The four themes covered in this subject are gathering useful data, turning data into information, probability and from data to decision making.

SUBJECT INTENDED LEARNING OUTCOMES (SILOs)

Upon successful completion of this subject, you should be able to:

- 1 Critique data-based conclusions that are reported in the media and similar outlets.
- 2 Interpret/derive simple numeric and graphical statistical summary measures of data.
- 3 Calculate probabilities in a variety of scenarios that may be used for informed decision making.
- 4 Identify and discuss common misconceptions of probability that can lead to poor decisions.

GRADUATE CAPABILITIES (GC)

GC	GC Point
Communication	Not evaluated in this subject
Inquiry and Analysis	Not evaluated in this subject
Personal and Professional	Not evaluated in this subject
Discipline Knowledge and Skills	Not evaluated in this subject

SELF-DIRECTED STUDY

In addition to the learning activities specified, you are expected to allocate approximately 4 to 5 hours per week of self-directed study in this subject. This time includes subject preparation, completing learning activities, readings, assessment preparation and other related work to enhance your participation in this subject and your learning at La Trobe.

LEARNING RESOURCES & TECHNOLOGIES

Resource Type	Title	Author and Year	Publisher
Required			
Subject Text	Data-Based Critical Thinking	Luke Prendergast, 2023	La Trobe

ASSESSMENT AND FEEDBACK SUMMARY

Note: all assessment instructions and rubrics with detailed criteria and level of performance are found on the Learning Management System two weeks prior to classes commencing.

Date due	%	Assessment	SILOs assessed
Official 1st semester exam period	60%	2-hour examination (equivalent to 2000 words)	ILO 1, 2, 3 and 4
Due no later than Friday 11.59pm in odd weeks starting from week 3	30%	Assignments (5 assignments, equivalent to 1500 words)	ILO 1, 2, 3 and 4
Quizzes must be completed no later than Friday 11.59pm in week 12	10%	Online numeracy skills quizzes (7 online quizzes, equivalent to 500 words)	ILO 2

Additional information about assessment & feedback

Students should keep a copy of all assignments submitted for assessment.

MANDATORY REQUIREMENTS

There are a variety of assessment hurdles or conditions for passing a subject other than the overall mark. **Hurdle Requirements** may be built into the assessment of the learning outcomes of your subject.

There are no Hurdle Requirements for this subject.

GENERAL ASSESSMENT REQUIREMENTS

SUBMITTING WRITTEN WORK WITH TURNITIN

Turnitin is a web-based text-matching software system used at La Trobe University to assist you in writing your assignments and checking for similarity with existing published work. Please remember to allow adequate time to submit your assignment to Turnitin. **It is your responsibility to have your assignment submitted by the assessment due date.** Not receiving an originality report is not an acceptable reason for requesting an assignment extension unless it is a required part of the assessment.

POLICIES, PROCEDURES AND GUIDELINES

The University has a comprehensive policy framework to which both staff and students must adhere. You should familiarise yourself with those policies, procedures and guidelines likely to affect you, especially the following:

- [Academic Progress Review Policy](#)
- Assessment Policy
- [Health and Safety](#) (OHS)
- [Privacy](#)
- [Student Academic Misconduct Policy](#)

ADJUSTMENTS TO ASSESSMENT

You may be affected by a range of adverse circumstances while you are preparing for or undertaking an assessment task. There are avenues for adjustments to your assessment depending on the circumstances you face including opportunities to apply for extensions to submission dates, Special Consideration and Alternative Assessment Schedules. For more detailed information refer to the [Assessment Procedure – Adjustments](#).

REQUEST AN EXTENSION OF TIME TO SUBMIT AN ASSIGNMENT TASK

Where you know in advance that you will not be able to submit an assessment task by the due date due to adverse circumstances that have affected you during the preparation of the task, you will need to [request an extension of time to submit](#). **This must be done at least three days prior to the due date.** Penalties normally apply if you submit an assessment after the due date. **Poor time management is not an acceptable reason for an extension.**

To apply, go to <https://www.latrobe.edu.au/students/admin/forms/request-an-extension/request>

SPECIAL CONSIDERATION

If you have experienced serious short term, adverse and unforeseen circumstances that substantially affect your ability to complete an assessment task to the best of your potential, you may be eligible to apply for **Special Consideration**. To do this or find more information, go to <http://www.latrobe.edu.au/special-consideration>

STUDENTS WITH A LEARNING ACCESS PLAN (LAP)

Students who have a Learning Access Plan must apply for any adjustments they may need via the same channels but may have access to more flexible arrangements depending on their circumstances. Please refer to the [Assessment Procedure – Adjustments](#) and your LAP.

REVIEW AND RE-MARK

For information on what to do if you wish to seek a review of an assessment outcome see Part B of the Validation and Moderation Procedure.

ACADEMIC INTEGRITY

Academic integrity means being honest in academic work and taking responsibility for learning the conventions of scholarship. Academic integrity education is integral to the learning experience at La Trobe University. The University requires its academic staff and students to observe the highest ethical standards in all aspects of academic work, and it demonstrates its commitment to these values by awarding due credit for honestly conducted scholarly work, and by penalising academic misconduct and all forms of cheating.

The penalty for submitting an assignment under your name that is the work of a third party may be severe, even leading to exclusion from the University without readmission. Refer to the **Academic Integrity - Schedule of Penalties and Actions** within the [Student Academic Misconduct Policy](#).

You should understand and be able to practice integrity in all your academic work, be familiar with the [Academic Integrity website](#) and complete the **Academic Integrity Module (AIM)** found in your LMS.

If you have any specific questions about **academic integrity** and your assessment, your Subject or Course Coordinator will be able to assist.

STUDENT FEEDBACK ON SUBJECT SURVEYS

The **Student Feedback on Subjects (SFS) Survey** is one of the quality assurance processes that occur across the University. In this survey, you can tell us about your learning experiences in this subject. We value your feedback. Your views are taken seriously and assist us to improve the learning experience for future students through the LTU Subject Monitoring process.

The surveys will be available for you to complete towards the end of the subject teaching period. Feedback from previous students who completed this subject is available in the LMS site for this subject.

For more information you may contact surveys@latrobe.edu.au.

CHARTER OF STUDENT RIGHTS AND RESPONSIBILITIES

As a La Trobe coursework student, you have rights and responsibilities which are governed by La Trobe statutes and regulations, as well as policies and procedures. La Trobe University aspires to provide an outstanding student experience which is conducive to scholarly activity and supportive of your learning goals.

The **Charter of Student Rights and Responsibilities** is a summary of your rights and responsibilities as a student at La Trobe. The Charter was developed in consultation with La Trobe students.
<https://policies.latrobe.edu.au/document/view.php?id=225>.

STUDENT COMPLAINTS

The **Student Complaints Management System (SCM)** is a formal mechanism for students to lodge concerns about any aspect of their university experience. A student complaint may be about an administrative matter, a problem with a subject or course, or the conduct of a student or staff member.

For more information about the **Student Complaint process** and how to lodge a complaint, visit the **Complaints and Feedback** page at: <https://www.latrobe.edu.au/students/support/complaints>.

GETTING HELP AND SUPPORT AT UNI

A range of services at La Trobe can assist you with your studies, health and wellbeing. For the full range of services and facilities available for you, go to **Student Help and Support** at <http://www.latrobe.edu.au/students/support>.

STUDYING ONLINE

When you are in studying an online subject, please be sure that you can continue to access our learning support services through the Learning Hub (academic skills, English language, Maths and Science) via Zoom appointments that you can book by accessing the [Learning Hub LMS page](#) or by emailing the LearningHub@latrobe.edu.au.

You also have access to assignment feedback and online study support chat through Studiosity that can be accessed from any subject [LMS](#) page. You can also access the Library services remotely through [the library website](#).

LIBRARY

Your **Library** offers many ways to support your study. To discover more about how the Library can help, visit your **Library website** and explore the following essential resources:

- **Achieve@Uni**
<https://www.latrobe.edu.au/students/study-resources/learning/achieve> - Learn skills to help complete your assessments
- **Assessment Planner**
<http://www.latrobe.edu.au/library/assessment-thesis-support/plan-my-assessment> – Create a personal assessment schedule
- **Expert Help Guides**
<https://latrobe.libguides.com/> - Find expert discipline and specialist help for your research
- **Referencing at La Trobe University**
<http://latrobe.libguides.com/referencing/> - Understand how and why to reference
- **Academic Referencing Tool**
www.lib.latrobe.edu.au/referencing-tool - Get detailed examples of referencing styles
- **Chat to library staff**
www.latrobe.edu.au/library - Access the text chat service from the Library website
- **Contact us**
www.latrobe.edu.au/library/about/contact - Call or email your Library

ASK LA TROBE

ASK La Trobe is a service open to all La Trobe students who need guidance with administrative matters during their time at La Trobe. ASK La Trobe can provide you with help relating to enrolments and timetabling, fees and financial support, exams, results and graduations & student life and support services.

Go to **ASK La Trobe – Current Students** at <http://latrobe-current.custhelp.com/app/home> and start by looking at our **FAQs** or visit us at your campus **ASK La Trobe Help Zone**.

If you can't find what you're looking for:

- **Call** 1300 La Trobe (1300 528 762)
- **Ask a question** online at <http://latrobe-current.custhelp.com/app/ask>
- **Visit the ASK La Trobe Help Zone.** Join the queue for the Bundoora, Bendigo or city campuses using the **QLess: Virtual Queuing system**.

LEARNING AND SUPPORT PROGRAMS

There are many resources, including services, to help you with your academic studies at La Trobe. Go to the **Learning and Support Programs** page at <http://www.latrobe.edu.au/students/study-resources/learning> for further information.

HEALTH AND WELLBEING

Your health and wellbeing are an important part of being a successful student. Studying can be stressful and many factors can affect your mental and physical health, in and outside of the University. Check out the services available to you at La Trobe to support your health and wellbeing at www.latrobe.edu.au/wellbeing.

WEEK-BY-WEEK OUTLINE OF TOPICS

Theme	Key Questions	Week	Lectures	Computer Lab
Gathering useful data	1. Does evidence collected from data effect everyday life? 2. Is the sample representative of the population of interest?	1	- Introduction to data - Introduction to critical thinking with data	
		2	- Populations and samples - Population parameters and estimates - Representative samples - Sampling bias and non-response bias - Simple random sampling	- Introduction to Excel 'N choose n' calculations - Proportions and percentages - Sampling
Turning data into useful information	1. How can we numerically and graphically explore the characteristics of variables? 2. How can we numerically and graphically explore the relationship between two variables?	3	- Understanding general formulae - Measures of location: Average and median - Proportions and percentages - Measures of spread: Range, variance and standard deviation	- Location and spread estimates in Excel - Using 'IF' statements in Excel to count occurrences
		4	- Graphical representations of data: Histograms, line plots, pie charts & column charts - Thinking critically with numerical and graphical measures	- Histograms and line plots in Excel - Symmetric and skewed histograms
		5	- Exploring relationships between two variables - Linear equations - Scatterplots and the correlation coefficient	- Scatterplots in Excel - Correlation and the importance of graphically assessing linearity
		6	- Comparing two categorical variables - Correlation does not imply causation - Confounding variables	Mid-semester revision
Uncertainty and probability	1. What is the relative frequency interpretation of probability? 2. What are the basic rules of probability?	7	- Observational studies and experiments - Relative frequency interpretation of probability - Equally likely outcomes	- Equally likely outcomes - Random number generation in Excel - Probability simulations
		8	- Important probability definitions and results - Events and complements of events	- Estimating probabilities from simulations in Excel - Relative frequency interpretation of probability and comparisons with estimated proportions
		9	- Probability and independent events - Dependent events and conditional probability	Calculating conditional probabilities
From data to decision	1. How can misconceptions of probability lead to flawed judgements? 2. What is the gambler's fallacy? 3. How can I use confidence intervals so that I, and others, can make informed decisions?	10	- Misconceptions of probability and flawed judgements - Coincidences - Simpson's Paradox - The gambler's fallacy	- Interactive computer test for ESP – coincidence or proof - An example of Simpson's Paradox
		11	- Confidence intervals - Confidence intervals for proportions - Confidence intervals for averages	- Simulating confidence intervals in Excel to understand 'confidence' - Decisions from confidence intervals
Revision		12	Revision	Revision